

Leon Riccius | Curriculum Vitae

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leon-riccius

Computational scientist (PhD candidate, TU Delft) building sampling methods and machine learning surrogates that make Bayesian inference methods affordable for expensive computational models. Experience across probabilistic ML (Gaussian processes, MCMC, active learning), deep learning, and differentiable FEM/CFD solvers, delivered as open-source Python (JAX, PyTorch) and C++ code.

Skills

Probabilistic ML & UQ:	Bayesian inference, MCMC (HMC/NUTS, PDMP samplers), Gaussian processes, active learning, surrogate modelling, hierarchical models
Deep Learning:	PyTorch, JAX, physics-informed neural networks, VAEs
Simulation:	Finite element method, solid mechanics, fluid dynamics
Engineering:	Python, C++, git, CI/CD, Linux/bash, AI-assisted development

Experience

Delft University of Technology **Delft, NL**
Doctoral Researcher *07/2021 – present*

- Developed a surrogate-assisted framework for piecewise-deterministic Markov process samplers (Zig-Zag, Bouncy Particle) to calibrate material model parameters.
- Outperformed tuned Random Walk Metropolis and No-U-Turn samplers on every metric; in 10 dimensions, $\sim 5\times$ lower posterior-mean error than NUTS at an equal computational budget.
- Calibrated a model of the weak interface zone in concrete from micro-scale displacement data: a 39-dimensional hierarchical Bayesian model over 10 specimens with a model-discrepancy term, on a GPU-accelerated differentiable FEM solver (JAX-FEM).
- Released the methods as open-source libraries, published my findings in peer-reviewed journal, and presented them at 8 international conferences, including NeurIPS (workshop) and SIAM.

Education

Delft University of Technology **Delft, NL**
Doctor of Philosophy, Civil Engineering *07/2021 – present*
Thesis: Piecewise Deterministic Markov Processes for PDE-Constrained Bayesian Inference

Technical University of Munich **Munich, GER**
Master of Science, Mechanical Engineering *03/2018 – 05/2021*
Thesis: Machine Learning Augmented Turbulence Modelling for the Reynolds Stress Closure Problem

Technical University of Munich **Munich, GER**
Bachelor of Science, Mechanical Engineering *10/2013 – 02/2018*
Semesters abroad: Keio University, Tokyo, JP (2019); Ecole Polytechnique de Montréal, CA (2016)

Selected Projects

pdmp: Library for Bayesian inference with piecewise-deterministic Markov process samplers, using cheap surrogates to propose events that are verified against the exact model (PyTorch, differentiable FEM in JAX).

mcmc-surrogates: Bayesian calibration of FEM model parameters where the expensive likelihood is replaced by an actively trained Gaussian process (C++).

pitbnn: Physics-informed tensor basis neural network for RANS turbulence closure, coupled to OpenFOAM

Teaching & Mentoring

TU Delft **Delft, NL**

Teaching assistant, course developer, and thesis supervisor *07/2021 – present*

- Co-supervised an MSc thesis with Airbus on surrogate modelling for uncertainty propagation (2024).
- Built interactive textbooks and taught courses on data science, AI, statistical learning, and uncertainty analysis for engineers.

Selected Publications

[1] Leon Riccius, Atul Agrawal, and Phaedon-Stelios Koutsourelakis. Physics-Informed Tensor Basis Neural Network for Turbulence Closure Modeling. In *37th Conference on Neural Information Processing Systems, NeurIPS workshop*, 2023.

[2] Leon Riccius, Iuri Rocha, Joris Bierkens, Hanne Kekkonen, and Frans van der Meer. Integration of Active Learning and MCMC Sampling for Efficient Bayesian Calibration of Mechanical Properties. *European Journal of Mechanics A/Solids*, 2026.

[3] Leon Riccius, Iuri Rocha, Joris Bierkens, Hanne Kekkonen, and Frans van der Meer. Piecewise Deterministic Markov Processes for Bayesian Inference of PDE Coefficients. 2026. under review.

Presented at 8 international conferences, including NeurIPS, SIAM AN, ECCOMAS, and UNCECOMP. Full list at leonrccs.github.io.

Awards & Languages

Awards: Travel Award, SIAM (2025); Best Poster Award, 27th Engineering Mechanics Symposium, 114 submissions (2024)

Languages: German (native), English (full professional, TOEFL iBT 111), French and Dutch (limited working), Japanese (elementary)